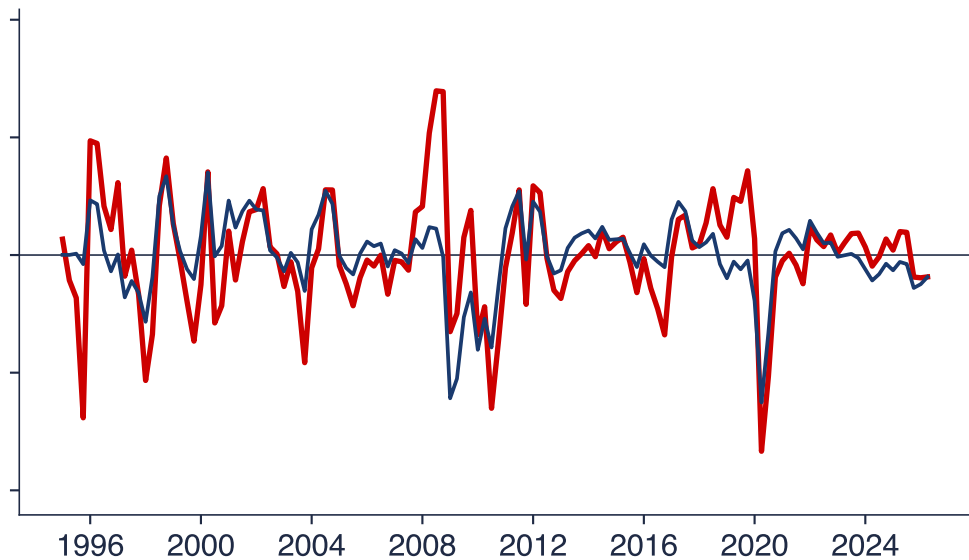
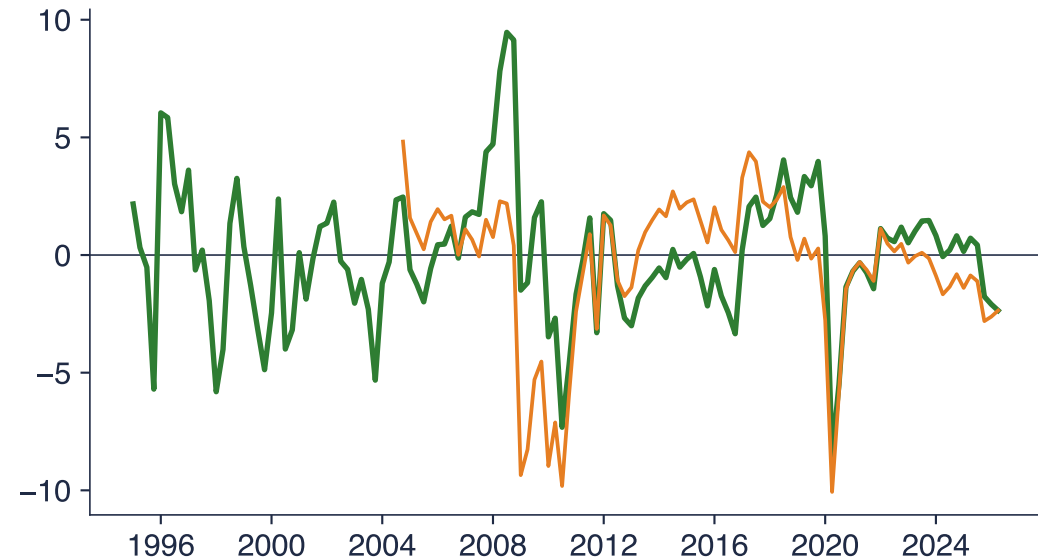


Hodrick-Prescott filter

unobserved-components model



— HP gap, final (two-sided)  
— HP gap, real time (one-sided)

— UC cycle, smoothed (two-sided)  
— UC cycle, filtered (one-sided)